SCST1143 – DATABASE ENGINEERING (2024/2025 - 01)

1. The following tables form part of a database held in a Relational Database Management System:

   Operator (**opCode**, opName)
   Journey (**opCode, destinationCode**, price)
   Destination (**destinationCode**, destinationName, distance)

   a. List the details of journeys less than £100.

   b. List the names of all destinations.

   c. Find the names of all destinations within 20 miles.

2. According to the tables 1, 2 and 3 below, please complete the instructions.

Table 1: Client

Cid	**Pname**	**Age**
5600	Harith Othman	36
6700	Lydia Lo	33
7011	Frodo Chandler	45
8099	Terry Oberoi	28

Table 2: Boat

Bid	**Bname**	**Color**
L201	Mutiara	White
S009	Jamaica's Eye	Blue
S223	Pacific	Yellow
L110	Tanjung	Red

Table 3: Order

Cid	**Bid**	**Date**
5600	L201	13/01/2024
7011	S223	25/02/2024
8099	L110	30/01/2024
5600	S009	10/02/2024

According to the tables above, answers the question in Algebra Relational.

i) List all the clients' information who are older than 30.

ii) Perform a Cartesian product between the Boat and Order tables.
iii) List all the boats and their details that were reserved after 30/01/2024.
iv) List the names of clients who have made a reservation.
v) List the names and colors of boats that have been reserved.
vi) List the names of clients who reserved boats with the color "Blue".
vii) Find the names of clients and the reservation dates for boats they reserved.
viii) List the details of orders for boats with the name "Pacific".